

# Package ‘carbonpredict’

August 18, 2025

**Title** Predict Carbon Emissions for UK SMEs

**Version** 0.0.4

## Description

This package uses pre-trained models to predict scope 1 and 2 carbon emissions for SMEs.

**License** MIT + file LICENSE

**Encoding** UTF-8

**Roxygen** list(markdown = TRUE)

**RoxygenNote** 7.3.2

**Imports** dplyr, ggplot2, progress, lmerTest

**Suggests** testthat (>= 3.0.0), mockery

**Config/testthat/edition** 3

**NeedsCompilation** no

**Author** Hamza Suleman [aut, cre, cph],  
Alec Phillpotts [ctb, aut],  
David Leake [ctb, aut]

**Maintainer** Hamza Suleman <Hamza.Suleman@lloydsbanking.com>

**Depends** R (>= 3.5.0)

## Contents

|                                   |          |
|-----------------------------------|----------|
| batch_predict_emissions . . . . . | 2        |
| batch_sme_plots . . . . .         | 2        |
| plot_sme_emissions . . . . .      | 3        |
| sme_emissions_profile . . . . .   | 4        |
| sme_scope1 . . . . .              | 4        |
| sme_scope2 . . . . .              | 5        |
| <b>Index</b>                      | <b>6</b> |

---

batch\_predict\_emissions

*Batch Predict Emissions*


---

### Description

Prediction entry point for batch SME and agriculture emissions

### Usage

```
batch_predict_emissions(data, company_type, output_path = NULL)
```

### Arguments

|              |                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data         | A single entry (list or named vector), a data frame, or a path to a CSV file. The data should contain company_name, 2-digit UK sic_code, and annual turnover columns. |
| company_type | A single parameter "sme" or "farm" to determine which emission prediction funtions to call.                                                                           |
| output_path  | Optional file path to save the results as a CSV. If NULL, results are not saved to a file.                                                                            |

### Value

A data frame with input columns and predicted emissions for each scope (in tCo2e). Optionally saved to a CSV file.

### Examples

```
sample_data <- read.csv(system.file("extdata", "sme_examples.csv", package = "carbonpredict"))
sample_data <- head(sample_data, 3)
batch_predict_emissions(data = sample_data, company_type = "sme", output_path = NULL)
```

---

batch\_sme\_plots

*Batch SME Plots*


---

### Description

Batch plot SME Scope 1 & 2 emissions

### Usage

```
batch_sme_plots(data, output_path = NULL)
```

### Arguments

|             |                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------|
| data        | A data frame or path to a CSV file with columns "sic_code", "turnover", and optionally "company_name". |
| output_path | Optional directory to save plots. If NULL, plots are not saved.                                        |

**Value**

Donut chart plots showing scope 1 and 2 predicted emissions (in tCo2e) for each row in the data. Optionally saved to a directory as PNG files.

**Examples**

```
sample_data <- read.csv(system.file("extdata", "sme_examples.csv", package = "carbonpredict"))
sample_data <- head(sample_data, 3)
batch_sme_emissions <- batch_predict_emissions(data = sample_data, company_type = "sme", output_path = NULL)
batch_sme_plots(data = batch_sme_emissions, output_path = NULL)
```

---

|                    |                           |
|--------------------|---------------------------|
| plot_sme_emissions | <i>Plot SME Emissions</i> |
|--------------------|---------------------------|

---

**Description**

Plot a donut chart of Scope 1 and Scope 2 emissions

**Usage**

```
plot_sme_emissions(scope1_emissions, scope2_emissions, company_name = NULL)
```

**Arguments**

|                  |                                                                         |
|------------------|-------------------------------------------------------------------------|
| scope1_emissions | Value for Scope 1 emissions (numeric).                                  |
| scope2_emissions | Value for Scope 2 emissions (numeric).                                  |
| company_name     | Optional company name to include in the chart title (character string). |

**Value**

A ggplot2 donut chart.

**Examples**

```
scope_1 = sme_scope1(85, 12000000)
scope_2 = sme_scope2(85, 12000000)
plot_sme_emissions(scope1_emissions = scope_1$predicted_emissions, scope2_emissions = scope_2$predicted_emis
```

---

sme\_emissions\_profile *SME Emissions Profile*


---

### Description

Calls the Scope 1 and Scope 2 emissions prediction functions and returns their results as a list and plots a donut chart

### Usage

```
sme_emissions_profile(sic_code, turnover, company_name = NULL)
```

### Arguments

|              |                                                              |
|--------------|--------------------------------------------------------------|
| sic_code     | A 2-digit UK SIC code (numeric).                             |
| turnover     | Annual turnover value (numeric).                             |
| company_name | Optional company name for labeling plots (character string). |

### Value

A list with two elements: scope1 and scope2, each containing the predicted emissions data frame (in tCo2e), as well as a donut chart.

### Examples

```
sme_emissions_profile(sic_code = 85, turnover = 12000000, company_name = "ABC")
```

---

sme\_scope1 *Predict SME Scope 1 Emissions*


---

### Description

This function loads a pre-trained emission model to predict scope 1 carbon emissions for a given SIC code and turnover.

### Usage

```
sme_scope1(sic_code, turnover)
```

### Arguments

|          |                                  |
|----------|----------------------------------|
| sic_code | A 2-digit UK SIC code (numeric). |
| turnover | Annual turnover value (numeric). |

### Value

A data frame with predicted emissions (in tCo2e) and input variables.

### Examples

```
sme_scope1(sic_code = 85, turnover = 12000000)
```

---

sme\_scope2

---

*Predict SME Scope 2 Emissions***Description**

This function loads a pre-trained emission model to predict scope 2 carbon emissions for a given SIC code and turnover.

**Usage**

```
sme_scope2(sic_code, turnover)
```

**Arguments**

|          |                                  |
|----------|----------------------------------|
| sic_code | A 2-digit UK SIC code (numeric). |
| turnover | Annual turnover value (numeric). |

**Value**

A data frame with predicted emissions (in tCo2e) and input variables.

**Examples**

```
sme_scope2(sic_code = 85, turnover = 12000000)
```

# Index

`batch_predict_emissions`, [2](#)

`batch_sme_plots`, [2](#)

`plot_sme_emissions`, [3](#)

`sme_emissions_profile`, [4](#)

`sme_scope1`, [4](#)

`sme_scope2`, [5](#)